



Exploring the design of AR-based procedural guidance tools to support reproducible laboratory workflows

Yuxin Shen^{1,2} · Wei Liang^{1,2} · Jianzhu Ma^{3,4} · Yixin Zhu^{5,6,7,8,9}

Received: 3 April 2026 / Accepted: 10 July 2026
© China Computer Federation (CCF) 2026

Abstract

Laboratory protocols serve as essential procedural guidance tools that support consistent outcomes and scientific innovation, yet they often fall short of facilitating reproducible results in real-world applications. Current methods for enhancing reproducibility in routine task protocols do not readily translate to laboratory experiments, due to their high procedural complexity and low tolerance for error. Inspired by the potential of Augmented Reality (AR) for procedural guidance, this paper explores the design of AR-based tools to support reproducible laboratory workflows. We conducted semi-structured interviews and participatory design sessions, using a lightweight AR prototype as a probe, to examine current protocol practices and challenges, and uncover experimenters' procedural information needs. Our findings reveal experimenters' expectations for the role of AR in experimental workflows and their concerns about its use. Based on these insights, we identify design opportunities for AR-based procedural guidance tools that better align with experimenters' needs and expectations, with the potential to enhance experimental efficiency.

Keywords Laboratory protocols · Empirical study · Participatory design · Augmented reality

1 Introduction

Laboratory protocols (i.e., procedural guidance tools) provide step-by-step instructions for conducting scientific experiments in fields such as biology (Selwyn 1996). They are essential for ensuring consistent experimental outcomes and serve as valuable knowledge resources for future generations of researchers (Hinkelmann and Kempthorne 2007;

O'Shaughnessy 2012). However, despite their importance, laboratory protocols often fail to yield reproducible outcomes in real-world practice, leading to reduced experimental efficiency and substantial resource waste (Baker 2016; Nagra et al. 2019). This gap can be attributed to two key characteristics of laboratory experimentation. First, laboratory procedures are often lengthy and complex, requiring the precise coordination of multiple parameters at each step,

✉ Wei Liang
liangwei@bit.edu.cn

Yuxin Shen
shenyuxin@bit.edu.cn

Jianzhu Ma
majianzhu@tsinghua.edu.cn

Yixin Zhu
yixin.zhu@pku.edu.cn

¹ School of Computer Science and Technology, Beijing Institute of Technology, 100081 Beijing, China

² State Key Laboratory of General Artificial Intelligence, Beijing Institute for General Artificial Intelligence, 100080 Beijing, China

³ Department of Electronic Engineering, Tsinghua University, 100084 Beijing, China

⁴ Institute for AI Industry Research (AIR), Tsinghua University, 100084 Beijing, China

⁵ School of Psychological and Cognitive Sciences, Peking University, 100871 Beijing, China

⁶ Institute for Artificial Intelligence, Peking University, 100871 Beijing, China

⁷ State Key Laboratory of General Artificial Intelligence, Peking University, 100871 Beijing, China

⁸ Wuhan Institute for Artificial Intelligence, Peking University, 430075 Wuhan, China

⁹ Beijing Key Laboratory of Brain-Computer Interface and Mental Health Modulation, Peking University, 100871 Beijing, China

including experimental conditions, reagent volumes, reaction times, and equipment settings (Nagra et al. 2019). Second, laboratory experiments are highly sensitive to errors, as even minor mistakes can significantly affect experimental outcomes (Gautier 2022). Therefore, developing a comprehensive understanding of the challenges experimenters face when using laboratory protocols in practice is essential for supporting more effective experimental work.

Similar issues arise in routine task protocols, such as cooking recipes (Ohsugi and Matsushita 2017) and assembly guides (Hayward 2025), which rely on sequential instructions yet can still lead to inconsistent outcomes. Prior research has attempted to improve the efficiency and usability of such protocols through textual refinements, including removing extraneous information (Leroy et al. 2013; Nisioi et al. 2017; Wu et al. 2024), simplifying complex material (Eom et al. 2012; Scarton et al. 2017; Al-Thanyyan and Azmi 2021), and segmenting lengthy procedures into more manageable steps (Chen et al. 2023). However, these strategies may not directly transfer to laboratory protocols. In routine tasks like cooking, skipping or reordering steps may still yield acceptable results, whereas even minor deviations in reagent volume or incubation time can invalidate an entire experiment. To improve procedural understanding and task performance, some approaches have explored more intuitive visual representations of procedural guidance, such as visual tutorials (Zhong et al. 2021) and video-based formats (Ouyang et al. 2024). Nevertheless, videos may fail to capture critical details in laboratory settings, particularly subtle techniques, safety cues, and context-specific decision points. Therefore, it is essential to systematically examine what information is most critical for experimenters and how guidance should be delivered to best support laboratory work.

Recent advances in AR technology have highlighted its potential to support task guidance and enhance user performance (Zhang et al. 2025; Ren et al. 2025). Prior work has applied AR to routine task domains such as cooking (Castelo et al. 2024; Wu et al. 2024; Zhai et al. 2020), assembly (Deshpande and Kim 2018; Alves et al. 2019; Funk et al. 2016; Hou et al. 2013; Lampen et al. 2019; Yang et al. 2019), and maintenance (Fiorentino et al. 2014; Obermair et al. 2020; Wang et al. 2022). In these contexts, AR presents task-relevant information within users' field of view, enabling users to stay focused on execution and reducing the need to constantly shift attention away from the workspace (Polvi et al. 2017; Buchner et al. 2022). Moreover, AR enables the flexible and situated presentation of procedural information through diverse modalities and cues (e.g., pointers, hand gestures, and in-context visual annotations), which has been shown to reduce cognitive load (Lai et al. 2019; Keelawat and Suzuki 2024), lower error rates (Castelo et al.

2024; Wu et al. 2024), and foster more engaging learning experiences (Weerasinghe et al. 2022; Zhu et al. 2023; Shi et al. 2025). Building on this line of work, we investigate how AR can better support laboratory procedures by examining experimenters' information needs and deriving design considerations for AR-based procedural guidance tools.

To this end, we investigate the following three research questions:

- **RQ1:** What current practices and challenges do experimenters encounter when using existing procedural guidance tools during laboratory workflows?
- **RQ2:** What information do experimenters need most during laboratory experiments, and how should procedural guidance be delivered to best support laboratory work?
- **RQ3:** What design opportunities emerge for AR-based procedural guidance tools to better meet experimenters' needs and support laboratory workflows?

To answer these RQs, we utilized a biology laboratory protocol as a representative case and conducted a two-phase study consisting of an exploratory interview study followed by participatory design sessions. In the first phase, we carried out semi-structured interviews ($N = 15$) to understand experimenters' current practices and pain points when working with existing laboratory protocols. We then used initial insights from these interviews to develop an AR prototype probe. In the second phase, we conducted participatory design sessions with this AR prototype probe to further examine experimenters' procedural information needs, their expectations for the role of AR in experimental workflows, and their concerns about its use. Building on these findings, we identify key design opportunities for AR-based procedural guidance tools that better align with experimenters' expectations, with the potential to improve experimental efficiency, reduce execution errors, and support more reproducible laboratory workflows.

In summary, we make the following contributions:

- We examine experimenters' current practices and challenges in using traditional laboratory protocols from an exploratory interview study.
- We provide an in-depth understanding of the procedural guidance information experimenters need to support laboratory work effectively.
- We explore the potential of AR to support laboratory workflows and derive a set of design opportunities for AR-based procedural guidance tools.

2 Related work

2.1 Laboratory assistance tools in the life sciences

Laboratory assistance tools have increasingly leveraged emerging technologies to accelerate scientific workflows and improve research productivity in the life sciences (Bhardwaj et al. 2022). Many studies leverage predictive modeling to automate labor-intensive scientific reasoning. For example, Shan et al. (2022) applied deep learning to predict protein 3D structures, while Baek et al. (2021) integrated multi-level sequence and spatial information for accurate protein folding prediction. Some researchers also explored Artificial Intelligence (AI)-assisted experimental planning to streamline scientific workflows. Roohani et al. (2024) proposed an AI-driven agent to support the planning of genetic perturbation experiments, and Qu et al. (2025) integrated an Large Language Model (LLM)-based agent with domain knowledge to guide gene-editing design workflows. Another direction focused on extracting procedural knowledge from scientific literature to facilitate automation and interpretation. For instance, Dannenfelser et al. (2023) developed structured knowledge representations extracted from biomedical literature, and Jiang et al. (2024) leveraged LLMs to translate scientific papers into intermediate representations for improved curation. In this paper, we extend this line of work by centering on hands-on experiment execution and investigating how laboratory procedural guidance tools can be designed to better support experimenters' procedural work.

2.2 Reproducibility challenges in laboratory work

While laboratory protocols aim to standardize procedures and support reproducible outcomes (Ioannidis 2015), reproducibility remains a significant challenge (Baker 2016). Over 70% of researchers report being unable to replicate others' results, wasting resources and eroding public trust in scientific research (Macleod et al. 2014). Two factors commonly contribute to these failures. First, the growing volume and complexity of scientific data (Plant and Hanisch 2020), combined with incomplete or underspecified protocols (Haddaway and Verhoeven 2015; Rahal et al. 2022), make procedures difficult to reproduce reliably. Second, scientific experiments are highly sensitive to error, and minor operational deviations or cognitive biases during hands-on execution can lead to inconsistent outcomes (Gautier 2022; Nagra et al. 2019).

To address these challenges, prior work has developed tools to improve protocol understanding and support accurate execution. For example, Liu et al. (2025) leveraged LLMs to support protocol reasoning, helping users better

interpret and follow procedural steps. Cui et al. (2024) introduced an intelligent monitoring system for tracking experimental processes. Despite these advances, there is a limited understanding of experimenters' information needs and how procedural guidance should be designed to support laboratory workflows effectively. Therefore, we address this gap by adopting a user-centered design approach to investigate experimenters' needs and inform the design of procedural guidance tools.

2.3 Enhancements for routine task protocols

Researchers have explored a range of approaches to improve routine task protocols and support more efficient task execution. A common direction focuses on refining protocol text to make instructions clearer and more concise (Siddharthan 2014). These methods leverage machine learning techniques to (1) replace complex terms with simpler alternatives (Carroll et al. 1999; Leroy et al. 2013), (2) restructure sentences to improve coherence and readability (Siddharthan 2006; Scarton et al. 2017; Al-Thanyyan and Azmi 2021), (3) remove redundant information (Nisioi et al. 2017), and (4) clarify complex concepts (Eom et al. 2012). Beyond text rewriting, Chen et al. (2023) used character recognition to decompose textual procedures into discrete actions, enabling more structured retrieval of step-level information. More recently, advances in LLMs have further automated and expanded text simplification and refinement. For example, Wu et al. (2024) demonstrated that GPT-3 can enhance task instructions by improving content, grammar, vocabulary, and overall clarity.

In parallel, researchers have explored more intuitive visual representations of procedural guidance to complement text and improve information accessibility (Agrawal et al. 2011; Leake et al. 2020; Xia et al. 2020). For example, Agrawal et al. (2011) proposed an approach to augmenting textbook sections with relevant web images, while Leake et al. (2020) developed a system that transforms textual descriptions into audiovisual content. Other work has focused on converting text-based procedures into alternative formats such as slide decks or video tutorials (Chi et al. 2021; Zhong et al. 2021; Ouyang et al. 2024). Chi et al. (2021) presented an automated system that converts text-based instructions into interactive videos with synthesized narration. Similarly, Zhong et al. (2021) introduced an automatic tutorial generation system that transforms text instructions into visual tutorials, improving task execution reliability.

While these methods can improve efficiency for routine protocols, they may not transfer seamlessly to laboratory experimentation. Routine tasks often take place in simpler and more forgiving contexts, whereas laboratory protocols

involve stricter setup requirements and greater procedural complexity. Moreover, laboratory work has a low tolerance for error, where even minor deviations can invalidate an entire experiment. Therefore, our research aims to examine how experimenters currently use laboratory protocols and what information they need during hands-on execution, in order to inform the design of more effective procedural guidance tools that better support laboratory workflows.

2.4 Potential of AR wearable devices to improve task efficiency

AR wearable devices (e.g., Apple Vision Pro, Meta Quest Pro) have become increasingly accessible and intelligent, enabling broader adoption in routine tasks such as cooking (Castelo et al. 2024; Wu et al. 2024; Zhai et al. 2020), assembly (Deshpande and Kim 2018; Alves et al. 2019; Funk et al. 2016; Hou et al. 2013; Lampen et al. 2019; Yang et al. 2019), and maintenance (Fiorentino et al. 2014; Obermair et al. 2020; Wang et al. 2022; Ren et al. 2025). By overlaying digital information directly onto the physical environment, AR reduces the need to shift attention between paper-based instructions and the workspace (Polvi et al. 2017; Buchner et al. 2022; Chen et al. 2023). For example, Chen et al. (2023) translated paper-based recipe instructions into an AR experience that provides contextualized guidance around task-relevant equipment. Wu et al. (2024) proposed an AR system that displays simplified task instructions in an AR headset, enabling hands-free access to procedural guidance. In addition, AR supports situated visual presentation of procedural guidance through diverse modalities and cues (Huang et al. 2021; Zhao et al. 2025). Huang et al. (2021) presented an adaptive AR-based task-tutoring system that employs rich visual elements, such as avatars, animated components, arrows, and step- and subtask-level expectations, to enhance user comprehension and reduce errors. Similarly, Zhao et al. (2025) proposed a

fully automated AR system that generates multiple forms of visual guidance from step-by-step instructions, enabling personalized learning experiences.

Collectively, these capabilities enable immersive systems to lower cognitive load (Lai et al. 2019; Keelawat and Suzuki 2024), foster greater learner engagement (Weerasinghe et al. 2022; Zhu et al. 2023; Shi et al. 2025; Anwer and Zagrouba 2025), and lead to improved task performance (Zhang et al. 2025; Ren et al. 2025). Motivated by these advances in routine tasks, we investigate how AR can support laboratory experimental procedures by examining experimenters' procedural information needs and deriving design considerations for AR-based procedural guidance tools that better support laboratory workflows.

3 Method

We conducted a two-phase study to examine the design of AR-based procedural guidance tools from a human-centered perspective, using biology as a representative domain. As shown in Fig. 1, the study comprised an exploratory interview study followed by participatory design sessions. In Phase 1, we conducted semi-structured interviews to understand current practices and the key challenges experimenters face when following existing laboratory protocols. The findings from this phase directly informed the design of an AR prototype probe. In Phase 2, we engaged participants in participatory design sessions using this probe to gain deeper insights into experimenters' procedural information needs and to explore how AR could better support laboratory experimental workflows. All research procedures were reviewed and approved in accordance with the university's ethical guidelines.

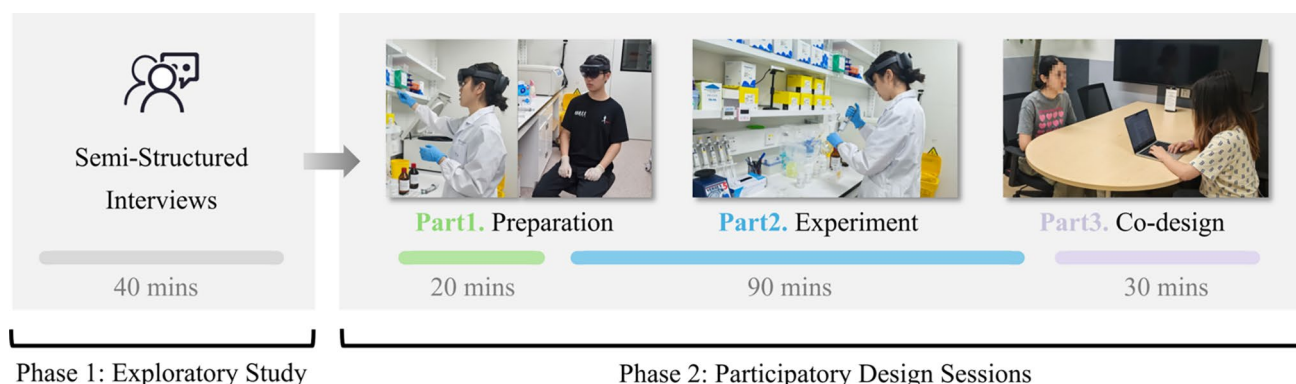


Fig. 1 Overview of the two-phase study. Phase 1 comprised semi-structured interviews with fifteen experimenters to identify current practices and challenges associated with existing laboratory protocols.

Phase 2 involved participatory design sessions using an AR prototype as a probe to investigate experimenters' expectations for AR-based procedural guidance tools

3.1 Data analysis

Data analysis incorporated audio and video recordings from the two phases. The initial transcription was performed using an automated system, followed by thorough manual verification by the authors. We employed an inductive thematic analysis methodology to identify themes from the collected data (Braun and Clarke 2006). The first two authors independently reviewed the transcripts to ensure a comprehensive understanding, followed by open coding procedures. These authors then engaged in two rounds of collaborative discussions to compare, categorize, and refine the generated codes, ultimately developing preliminary coding schemes and thematic groupings. During this process, whenever coding differences or inconsistencies arose, the authors revisited the original transcripts, compared alternative interpretations, and resolved disagreements through discussion and consensus. In addition, the remaining authors reviewed the analysis materials from both phases. For the Phase 2 prototype sessions, video recordings supplemented the transcript analysis by providing additional evidence of participants' behaviors, interactions, and preferences, and by enabling cross-checking with the corresponding transcripts. The entire author team participated in iterative discussions to evaluate and refine both the code and the themes, reaching unanimous consensus on the final thematic framework. This collaborative and iterative process helped improve coding consistency, reduce individual interpretive bias, and enhance the transparency and reproducibility of the final thematic framework.

Table 1 Demographic information of the participants in the semi-structured interview session

Participant	Gender	Age (years old)	Work experience (years)
P01	Male	26–30	5–6
P02	Female	26–30	3–4
P03	Male	31–35	3–4
P04	Female	31–35	3–4
P05	Female	26–30	7–8
P06	Female	21–25	5–6
P07	Female	26–30	9–10
P08	Female	21–25	2–3
P09	Female	21–25	1–2
P10	Male	21–25	3–4
P11	Female	21–25	1–2
P12	Female	21–25	2–3
P13	Female	21–25	4–5
P14	Female	26–30	4–5
P15	Female	26–30	5–6

4 Phase 1: Exploratory study

In Phase 1, we conducted semi-structured interviews to examine experimenters' current workflows and challenges when using existing laboratory protocols. The insights gained from these interviews directly informed the design of the AR prototype probes used in the Phase 2 participatory design sessions.

4.1 Semi-structured interviews

We conducted semi-structured online interviews with fifteen participants (3 males, 12 females) via Zoom, each lasting approximately 40 min. Participants were university students or research assistants, aged 21–33 years ($M = 26.53$, $SD = 3.78$). Detailed demographic information, including gender, age, and research experience, is provided in Table 1. Before each interview, we explained the purpose and procedures of the study and obtained informed consent from all participants. The interview protocol focused on three areas: (1) current practices in using existing procedural guidance tools, (2) the perceived role and importance of protocols in experimental work, and (3) challenges experimenters encounter when following existing procedural guidance. Example questions included “What types of procedural guidance tools do you typically use in your daily experiments?” and “Can you describe challenges you encountered when following existing procedural guidance tools in recent experiments?” All interviews were audio-recorded with participants' permission. Upon completion, participants received \$15 in compensation.

4.2 Interviews findings (Phase 1)

Participants demonstrated strong engagement throughout the exploratory interviews and emphasized the importance of thoroughly analyzing existing procedural guidance tools to enhance their experimental experience. As a result, they provided detailed insights into their routine experimental practices and the challenges they encounter.

4.2.1 Practices and perceived importance of laboratory protocols

All participants ($N = 15$) reported frequent use of paper-based protocols as references during their experiments. These protocols, whether provided in manufacturer kits or prepared in the laboratory, are typically placed on the lab bench and often covered with plastic wrap to protect them from contamination by reagents. Participants emphasized that these protocols are essential sources of experimental information, particularly when conducting unfamiliar

procedures. Even when performing well-practiced experiments, a majority of participants ($N = 10$) stated that they still referred to protocols for specific procedural details. Due to the complexity of certain tasks, such as adding multiple reagents, remembering the exact volume of each component was described as challenging. As one participant (P5) explained, *“I rely heavily on the paper-based protocols, even for experiments I have performed before. It is crucial for me to verify key details such as equipment settings and reaction times to ensure that all experimental steps are followed correctly and nothing is overlooked.”*

4.2.2 Key challenges in using laboratory protocols

The interviews also revealed three fundamental limitations that may affect how experimenters perform their experiments.

C1: The common paper-based format of protocols is inconvenient and cumbersome to use during experiments. Many participants ($N = 13$) reported that the paper-based protocol format posed practical challenges during experiments. Protecting paper protocols from spills and contamination complicates access to them. As P4 explained, *“We often cover the protocols with plastic wrap to keep them safe, but this makes it harder to read and flip pages.”* Additionally, handling paper while wearing gloves or working with instruments is inconvenient. P2 mentioned, *“When my hands are gloved or busy, it’s hard to handle paper. I often have to remove gloves or ask a colleague to help me find the next step.”*

C2: Textual descriptions in laboratory protocols are often vague and lack sufficient detail. Many participants ($N = 14$) reported that textual protocols alone were insufficient for effective experimental guidance. This limitation manifested in two primary ways. First, protocols often used unfamiliar or ambiguous terminology, hindering understanding. As participant P6 noted, *“Textual protocols often lack clarity in describing the state of reactants. I do not understand what a flocculent precipitate looks like.”* Second, certain operational details are often excluded or insufficiently communicated through textual descriptions in protocols. P3 emphasized this point, *“I often spend time searching for instructions on the required hand gestures or procedures for instruments during novice experiments.”*

C3: Experimenters with different levels of proficiency in a specific experiment may have distinct protocol needs. Our analysis suggests that protocol requirements may vary depending on an experimenter’s level of expertise. All participants ($N = 15$) emphasized the need for highly detailed protocols when conducting unfamiliar experiments. As P12 explained, *“When conducting a new experiment, I prefer the protocol to be highly detailed, as I lack experience with the*

procedure and am unfamiliar with the key points of each step.” In contrast, the same participants noted that such detailed protocols can be cumbersome when performing well-practiced procedures. As P5 described, *“When reading textual protocols for proficient experiments, I spend a lot of time searching for the key steps because the text is too lengthy.”*

5 Phase 2: Participatory design sessions

Building on insights from the interviews, we conducted participatory design sessions using an AR prototype as a probe to explore experimenters’ information needs and inform design opportunities for AR-based procedural guidance tools.

5.1 Prototype probe design

Based on the challenges (C1–C3) identified in Phase 1, we designed a lightweight AR prototype to deeply explore experimenters’ procedural information needs during experiment execution, incorporating the following features:

- *Providing accessible and hands-free procedural visualization (C1):* We adopted the Microsoft HoloLens 2 AR headset for its hands-free interaction, which preserved sterility and accessibility, and its eye-tracking and first-person recording capabilities for analyzing participants’ interactions with procedural content.
- *Integrating video tutorials with textual protocols (C2):* Building on prior research showing that visual information enhances contextual understanding and supports accurate task execution (Chi et al. 2021; Zhong et al. 2021), we incorporated video tutorials into traditional protocols to assess their impact on experimental performance.
- *Accounting for differences in experimenter expertise (C3):* We organized participants into two groups to examine how procedural information needs differ across users with varying levels of expertise. One group consisted of novice experimenters who were conducting the selected biological experiment for the first time, while the other group comprised experienced experimenters already familiar with the experimental workflow.

We integrated these three probe design considerations into a lightweight AR prototype probe, as illustrated in Fig. 2. We implemented the experimental platform in Unity and deployed it on the HoloLens 2 using the Mixed Reality Toolkit 3.¹ The system presented video tutorials alongside

¹ <https://learn.microsoft.com/en-us/windows/mixed-reality/mrtk-unity/mrtk3-overview/>.

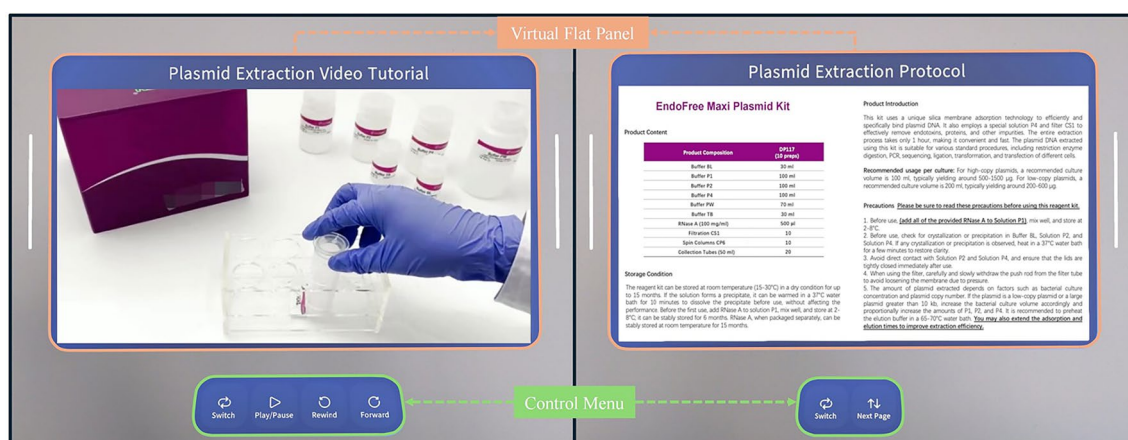


Fig. 2 Interface of the lightweight AR prototype probe. The plasmid extraction video tutorial (left) is displayed on a virtual panel with controls for playback, including play, pause, rewind, and fast-forward.

textual protocols on a configurable virtual panel, allowing users to flexibly switch between formats. Spatial anchoring and voice commands were incorporated to ensure that virtual content remained stable, hands-free, and easily accessible throughout the laboratory workflow.

5.2 Participants

We recruited participants through mailing lists from multiple universities affiliated with the authors. Participants were required to be familiar with laboratory safety regulations and proficient in using basic laboratory equipment (e.g., centrifuges, pipettes, and vortex mixers) to ensure they could safely and effectively perform the biological experiments. In total, we recruited four participants (age $M = 25.25$, $SD = 5.5$, two females, two males), including two novices and two experts. Novices had 1–3 years of laboratory experience, whereas experts had more than three years of experience. None of the participants had prior experience using AR headsets. Before the experiment, all participants completed a familiarization and training session on the AR prototype, as described in Sect. 5.3.1. Each participant received \$50 upon completing the study.

5.3 Procedure

The session comprised three phases, including preparation, experiment, and co-design, totaling approximately 140-min, with the experiment conducted in an accessible laboratory. Prior to starting, each participant reviewed and signed an approved informed consent form detailing the study protocol and data collection methods.

The corresponding textual protocol (right) is presented in a similar format, featuring navigation controls for paging through procedural steps



Fig. 3 Preparation of all reagents and equipment. The materials include essential reagents, instruments, and key lab equipment such as a centrifuge, pipettes, and a vortex mixer

5.3.1 Preparation

We first conducted a 20-min preparation phase to familiarize participants with both the experimental protocol and the AR headset. Participants initially reviewed the procedure using video tutorials on smartphones and paper-based protocols. We then trained participants on fundamental AR interactions, including virtual object manipulation and control mechanisms. Participants practiced these interactions until they could operate the system smoothly, ensuring that system use did not interfere with experimental execution. All necessary reagents and equipment were prepared and arranged on the laboratory bench prior to the experiment, as shown in Fig. 3.

5.3.2 Experiment

During the experimental session, participants performed an Endo-free Plasmid Maxi Extraction. We selected plasmid

extraction as a representative task because it is a fundamental yet well-structured molecular biology procedure, and one of the few experiments accompanied by detailed video tutorials in addition to textual protocols. This made it well-suited for exploring multimodal procedural guidance. The experiment comprised key steps including bacterial culture, cell lysis, solution filtration, plasmid adsorption, purification, and collection. The session lasted approximately 90-min and was conducted using the AR guidance system (Fig. 4). We recorded continuous first-person video with integrated eye-tracking to capture participants' attention patterns and protocol use throughout the experiment. A scheduled 10-min break was included to mitigate fatigue from prolonged headset use.

5.3.3 Co-design

Following the experimental phase, we conducted 30-min co-design sessions in a separate meeting room using a retrospective think-aloud method (Lazar et al. 2017). Participants reviewed their first-person recordings with gaze visualizations, offering moment-by-moment reflections on their decisions and information preferences. Structured questions guided discussions on participants' experiences with textual protocols and video tutorials, their perceptions of AR display strategies, and which types of information they found most effective. Sessions were documented through audio recordings and detailed notes, yielding rich qualitative insights to complement our experimental data.



Fig. 4 Experiment with the AR prototype probe. Participants conducted the plasmid extraction procedure while following virtual guidance displayed through the HoloLens 2 AR headset

6 Findings (Phase 2)

In this section, we summarize findings from the participatory design sessions to examine how the AR prototype probe influenced experimenters' practices and to explore their procedural information needs during experimentation. The findings from this phase can be categorized into four areas: (1) the need for clearer descriptions and better-structured textual protocols, (2) the value of multimodal content in offering concrete demonstrations and complementary understanding, (3) the importance of tailoring protocols to different levels of expertise and cognitive preferences, and (4) the potential of interactive and accessible formats to enhance experimental efficiency. In summary, participants envisioned intelligent laboratory protocols as intelligible, rich media, adaptive, and accessible tools that address the complexity and rigor of real-world laboratory experiments. We summarize the Phase 2 findings in Fig. 5.

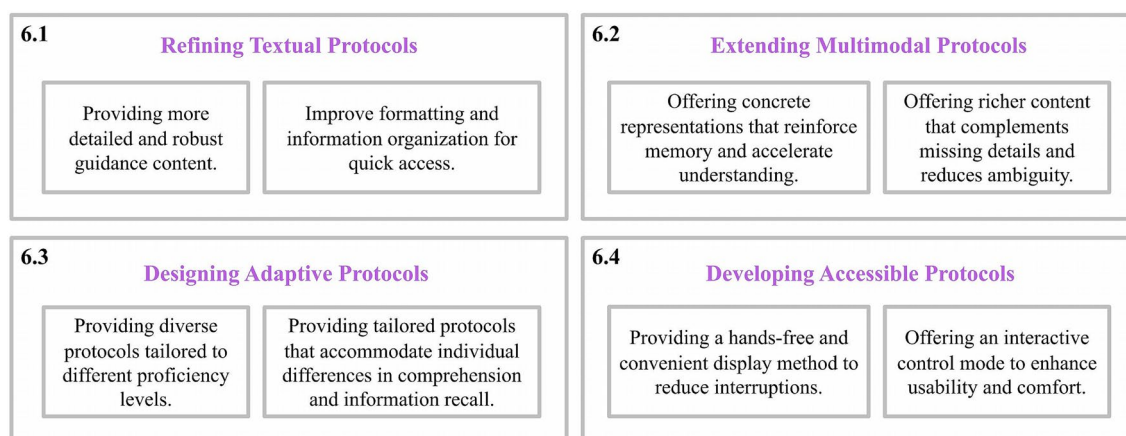


Fig. 5 Overview of findings from Phase 2. We summarize experimenters' expectations for improving existing protocols in four key directions: (1) refining textual protocols by enhancing content detail and organization, (2) extending multimodal protocols through concrete

representations and complementary information, (3) designing adaptive protocols that accommodate diverse user proficiencies and cognitive styles, and (4) developing accessible protocols with hands-free interaction and improved usability

6.1 Refining textual protocols to improve textual descriptiveness and information organization

All participants recognized textual protocols as vital sources of experimental information, including product specifications and operational steps. They particularly valued these protocols for outlining precautions and optional procedures that help them follow procedures with greater confidence. Nevertheless, participants emphasized the need to enhance both the clarity and comprehensiveness of the protocols' content and format.

6.1.1 Textual protocols should provide more detailed, robust guidance

Participants consistently highlighted issues of ambiguity and insufficient detail in existing protocols. A common challenge was the use of vague terminology, often relying on personal interpretation rather than standardized language. For instance, H03 remarked, *"The protocol instructs to invert the tube 6–8 times gently, but the meaning of 'gently' is unclear and subjective."* Moreover, protocols frequently provide abstract instructions that lack crucial procedural details or omit essential steps and instruments between adjacent procedures. H02 illustrated this concern by stating, *"An instruction to 'Add 8 ml of Buffer P2 to the centrifuge tube' omits critical steps such as ensuring the solution is at room temperature."*

Additionally, a significant gap emerged regarding guidance for unexpected circumstances. Participants noted that protocols rarely include instructions for real-time adjustments. For example, they often lack guidance on how to modify reagent quantities if initial measurements deviate from specifications. Environmental factors affecting reagent reactions, such as room humidity or dryness, are often unaddressed, leaving experimenters to resolve issues through trial and error. As H03 explained, *"Sometimes the buffer behaves differently depending on the room conditions, but the protocol doesn't say how to adjust for that, so I have to figure it out myself."* Therefore, providing more rigorous and specific descriptions is essential to enhance protocol clarity and reliability, thereby supporting accurate execution.

6.1.2 Textual protocols should improve formatting and information organization for quick access

Participants emphasized that a key issue with textual protocols is their suboptimal formatting and disorganized information structure, which hinders efficient access and comprehension during experiments. First, the extensive textual descriptions are overly dense, making procedural information harder to process and causing reading fatigue.

Experimenters must parse full sentences to locate essential information, slowing both preparation and execution. H02 expressed this frustration, *"These text-based protocols are so lengthy that I have to spend considerable time reading and understanding the procedures. During the experiment, finding key information amidst the dense text is also quite frustrating."*

Second, inconsistent emphasis on critical information further compounds the problem. Different content types, such as considerations, optional steps, and general explanations, are often presented using the same bold formatting. This overuse dilutes the visual distinction of truly essential actions, making it harder for experimenters to quickly identify what matters most. Under time pressure to meet reaction deadlines, they may inadvertently overlook crucial instructions. H04 noted, *"When conducting experiments, I focus on the key steps. If considerations are highlighted in bold rather than color, I tend to view them as explanations rather than essential information."*

Third, information relevant to a single procedural step is often scattered across different parts of the protocol, making it difficult to locate and follow. Critical considerations placed before or after numbered steps are frequently overlooked during execution. As H01 explained, *"I usually focus on the numbered steps and sometimes forget to check the bolded notes or considerations listed before the operation steps."* Moreover, the sequencing of steps often fails to align with the actual workflow. H04 pointed out, *"The protocols have issues with step order. For example, the first step should be performed alongside the sixth, yet they are not placed together."* In summary, improving the formatting and organization of textual protocols is essential to facilitate rapid and clear access to critical information and make procedural details easier to locate and interpret.

6.2 Extending multimodal protocols to provide concrete demonstration and complementary information

In the co-design session, participants expressed that integrating video tutorials improved their understanding and execution of the laboratory procedures.

6.2.1 Leveraging visual information to offer concrete representations that reinforce memory and accelerate understanding

Participants noted that video tutorials are especially effective in providing concrete visual demonstrations that support both material recognition and procedural memory. The appearance of experimental materials, such as reagents and instruments, is intuitively conveyed, allowing experimenters

to quickly identify and locate necessary items. As H04 shared, *“I had never done this kind of experiment before, so I wasn’t familiar with the materials, like the different types of tubes. Watching the video tutorial really helped because it clearly showed me which materials I needed to use.”* In addition, step-by-step visual demonstrations help reinforce procedural memory. H01 emphasized, *“Video tutorials clearly demonstrate the sequence of operations, and the visual cues are clearly displayed in my mind, so I can quickly recall how to perform each step. Cementing the steps in my mind helps me feel more confident performing them.”*

The progressive structure of video tutorials further enhances comprehension by presenting content systematically from an experimental introduction and preparation through procedures to common problems and solutions. This layered presentation supports gradual understanding and readiness. As H01 appreciated, *“Starting with an introduction and preparation gives me time to get ready and think things through.”* Moreover, the focused presentation of key procedural elements enables efficient identification of critical information. H04 noted, *“I prefer the video tutorial procedures because they focus on key details like equipment settings, reagent amounts, reaction times, and temperature settings, making it easy to find the key information I need quickly.”* H03 also highlighted the value of built-in troubleshooting guidance.

6.2.2 Leveraging multimodal information to offer richer content that complements missing details and reduces ambiguity

All participants consistently supported incorporating multimodal information, as it provides richer and more diverse perspectives. On the one hand, textual protocols complement video tutorials by including more detailed information, such as key considerations, experimental tips, and alternative procedures that may be omitted from videos. Video tutorials often prioritize the procedure’s primary steps and may omit important details. As H01 exemplified, *“The textual protocol states that if there’s too much biomass, you can extend the centrifugation time to separate the precipitate from the waste liquid better. However, this tip isn’t mentioned in the video tutorial.”*

On the other hand, video tutorials complement textual protocols by offering visual clarity for nuanced details that are difficult to convey through text alone. Some operational subtleties are difficult to describe, leaving experimenters unsure whether they are performing tasks correctly. H02 highlighted this gap: *“The textual protocols don’t tell me how to open the lid, or that the tip of the pipette shouldn’t be placed in the liquid. These details are so subtle that designers often don’t record them. However, I can easily find this*

information in the video tutorial.” Video demonstrations also help illustrate the physical states of reacting solids and liquids, providing crucial visual cues for correct interpretation. Together, textual and visual formats offer complementary strengths, and their integration supports clearer understanding and more accurate execution of laboratory procedures.

6.3 Designing adaptive protocols to accommodate expertise levels and cognitive preferences

The initial findings of Phase 1 indicated that experimenters with varying proficiency levels have distinct protocol requirements. Building on this, Phase 2 offered a deeper investigation into how these differences manifest in both content and format preferences, and uncovers the underlying causes.

6.3.1 Providing diverse protocols tailored to different proficiency levels

Our analysis indicated that experimenters’ procedural requirements develop progressively with increased experience, influencing both their preferred formats and the nature of content they seek. Clear progression patterns were observed in format preferences. Results showed that novice experimenters relied heavily on video tutorials, whereas experienced practitioners primarily consulted textual protocols. Interview data further indicated that video tutorials serve as accessible entry points when learning new procedures, while textual protocols have the potential to support refinement and efficiency as expertise develops. H04 described this developmental shift, *“When starting a new experiment, I rely heavily on video tutorials and text-based protocols. However, the video tutorials have become less important as I have become more familiar with them.”* These findings suggested that protocol formats should be adaptable, offering multiple representations to support users at different stages of expertise.

Content requirements also evolve in parallel with the development of expertise. Novice experimenters tend to prioritize a clear understanding of fundamental objectives and the accurate, efficient completion of procedures. H01 exemplified this early-stage need, *“At the beginning, I require the protocol to be detailed. I want to understand the purpose of each step and the precautions involved; otherwise, I lack confidence in performing the procedures correctly.”* As proficiency advances, experimenters increasingly focus on optimizing experimental outcomes by attending to nuanced protocol details. H04 emphasized, *“Having become familiar with the procedure, I now concentrate on the steps that*

influence reaction efficiency. I no longer read everything. Only the parts that could affect the results.”

Combining these observations with the protocol information categories defined by Selwyn (1996), we identified distinct content needs for novice and proficient experimenters. Novice experimenters require broader and more explanatory content, including detailed theoretical background, comprehensive precautions, and step-by-step rationales. In contrast, proficient experimenters primarily focus on material specifications and core operational details. Moreover, discussions revealed that proficient experimenters sometimes possess expertise that exceeds the scope of existing protocols, underscoring the need for adaptable, evolving protocol content.

6.3.2 Providing tailored protocols that accommodate individual differences in comprehension and information recall

Discussions with participants revealed notable individual differences in how experimenters comprehend, retain, and apply protocol information. While some experimenters quickly internalized new workflows by identifying key decision points, others relied on repeated consultation of protocol materials to build confidence and ensure procedural accuracy. For example, when learning a new protocol, H01 rapidly extracted and acted on critical steps, whereas H04 reported rereading each instruction multiple times to avoid mistakes.

Additionally, certain individuals display strong numerical recall and heightened sensitivity to quantitative parameters such as reagent volumes, reaction durations, and instrument settings. These experimenters benefit from protocols that foreground precise numerical values, allowing them to operate with minimal reference once core metrics are memorized. As H02 explained, *“I pay close attention to the exact measurements and timing. Once I have those memorized, I feel confident proceeding without rereading the protocol.”* These findings underscore the importance of designing protocol systems that are adaptable not only to users’ proficiency levels but also to their diverse cognitive preferences and modes of information engagement. Tailoring protocol design to individual cognitive characteristics may be perceived as improving procedural efficiency and user confidence across a wide range of users.

6.4 Developing accessible protocols to enable hands-free interaction and seamless access

All participants emphasized that accessible protocol formats offer substantial convenience during laboratory work. Traditional methods, such as referring to printed documents

or viewing protocols on mobile devices, were described as cumbersome, particularly when wearing gloves or handling sensitive materials. Participants noted that minimizing physical interaction with external devices is crucial for maintaining workflow continuity and preventing contamination. As H01 explained, *“If I use the HoloLens 2, once it’s set up, the protocol screen stays fixed in view. I just need to turn my head to see it. It’s always there when I need it.”* This hands-free, persistent AR display significantly enhances usability, allowing experimenters to reference instructions without interrupting their procedures.

In addition to hands-free accessibility, participants also emphasized the importance of interactivity in enhancing the usability of protocols. One notable advantage of AR-based systems is the seamless transition between different media formats. For example, switching between textual instructions and video demonstrations can be achieved with a simple virtual button press, eliminating the need for direct physical contact with surfaces or devices. Participants also highlighted the value of audio guidance, which allows for cognitive and physical decoupling during task execution. As H04 noted, *“Once you’re familiar with the procedure, you can quickly perform the manual steps while listening to the instructions. It feels very convenient.”* Moreover, the display capabilities of AR headsets significantly improve readability. Compared to mobile phones or printed documents, the virtual protocol interface can be scaled to a larger size with clearer text. As H03 described, this enhanced visual clarity reduces the need for constant adjustment or squinting, contributing to a more streamlined and comfortable user experience during complex procedures.

7 Discussion

Our study examined current practices and challenges associated with existing laboratory protocols, as well as the role of AR in supporting experimental work, underscoring the need for comprehensive, well-organized, multimodal, adaptive, and accessible procedural guidance tools. While prior work on procedural enhancement has primarily focused on everyday tasks (Semeraro and Vidal 2022; Chen et al. 2023; Wu et al. 2024), laboratory experimentation introduced greater complexity, stricter constraints, and lower tolerance for error. These characteristics call for domain-specific design considerations when applying AR to laboratory workflows. Below, we outline key design opportunities for advancing AR-based laboratory procedural guidance tools, grounded in cognitive theory and informed by both routine-task research and our empirical findings.

7.1 Design implications for AR-based procedural guidance

Using the lightweight AR prototype as a design probe, the participatory design sessions elicited situated feedback from experimenters on how AR-based procedural guidance could support laboratory workflows. As shown in Sect. 6, participants' feedback centered on four major needs: clearer and better-structured textual protocols, richer multimodal support, adaptive guidance for different expertise levels and cognitive preferences, and accessible interaction during hands-on laboratory work. These findings suggested that AR-based procedural guidance tools should not simply reproduce existing protocols in a new display format, but should carefully consider how procedural information is structured, represented, adapted to users, and accessed during hands-on laboratory work. We therefore discuss four design implications that connect these empirical findings with the design of future AR-based procedural guidance tools.

7.1.1 Balancing information density and cognitive load

As shown in Sect. 6.1, participants identified two main needs regarding textual protocols: more specific and robust procedural descriptions, and clearer organization of critical information. Existing textual protocols often required experimenters to interpret ambiguous information, infer missing details, and search through lengthy or scattered text for key parameters and cautions. These issues increased the burden of understanding, locating, and retaining procedural information, suggesting that AR-based procedural guidance systems should carefully balance information richness with cognitive accessibility. According to cognitive load theory, presenting highly detailed or fragmented information without appropriate structure can overwhelm users, especially in time-pressured and cognitively demanding contexts (Feinberg et al. 2000; Chandler and Sweller 1991). For AR-based procedural guidance tools, this highlights an opportunity to move beyond static protocol presentation toward a cognitively informed approach to information orchestration. Rather than displaying all details simultaneously, AR systems can dynamically reveal information at appropriate levels of granularity, which may support more manageable processing of procedural information while preserving necessary procedural details.

Although prior work has explored text simplification and restructuring for everyday tasks (Siddharthan 2006; Scarton et al. 2017; Al-Thanyyan and Azmi 2021; Nisioi et al. 2017; Wu et al. 2024), directly applying these techniques to laboratory protocols risks oversimplifying critical operational details. Instead, AR systems present an opportunity to

support structural refinement rather than content reduction. For example, experimental workflows can be segmented into meaningful phases (e.g., lysis, purification), each corresponding to a clear subgoal. This allows experimenters to focus on one phase at a time while maintaining awareness of overall progress. Such phase-based chunking aligns with cognitive principles by reducing unnecessary navigation and working-memory demands, and can support more accurate and reliable execution of complex laboratory procedures.

7.1.2 Integrating multimodal content and context-aware presentation

As shown in Sect. 6.2, participants emphasized that textual protocols and video tutorials provided complementary forms of procedural support. Video tutorials helped participants understand material appearances, operational sequences, and subtle techniques through concrete demonstrations, while textual protocols provided detailed cautions, optional procedures, and troubleshooting information that videos may omit. These findings suggest that AR-based laboratory procedural guidance tools should integrate multimodal information within the experimental workspace. By presenting procedural guidance through multiple channels such as text, visuals, and audio, AR systems can more effectively convey complex experimental information than text-only protocols. Consistent with the modality effect in cognitive psychology (Knoop-van Campen et al. 2018), distributing information across complementary modalities can reduce cognitive load and support comprehension and retention, particularly in structured and cognitively demanding laboratory tasks. Building on this principle, AR procedural guidance tools can incorporate diverse modalities, including images, audio narration, and interactive flowcharts, all displayed in situ within the AR environment (Fig. 6). Presenting these modalities via AR headsets enables experimenters to access rich procedural cues without diverting attention from ongoing experimental work. Such multimodal AR presentations can potentially improve communication clarity and accommodate variation in users' learning and cognitive preferences.

Because different information forms support different aspects of procedural understanding, their presentation should also be coordinated with the experimental context. This highlights the importance of context-aware timing in AR-based procedural guidance. Drawing on learning principles and cognitive load theory (Govindasamy 2001), instructional support is most effective when delivered at the moment it is needed during the experimental workflow. AR systems are well-positioned to orchestrate the presentation of different modalities based on task context and execution state. For example, high-level overview materials such as

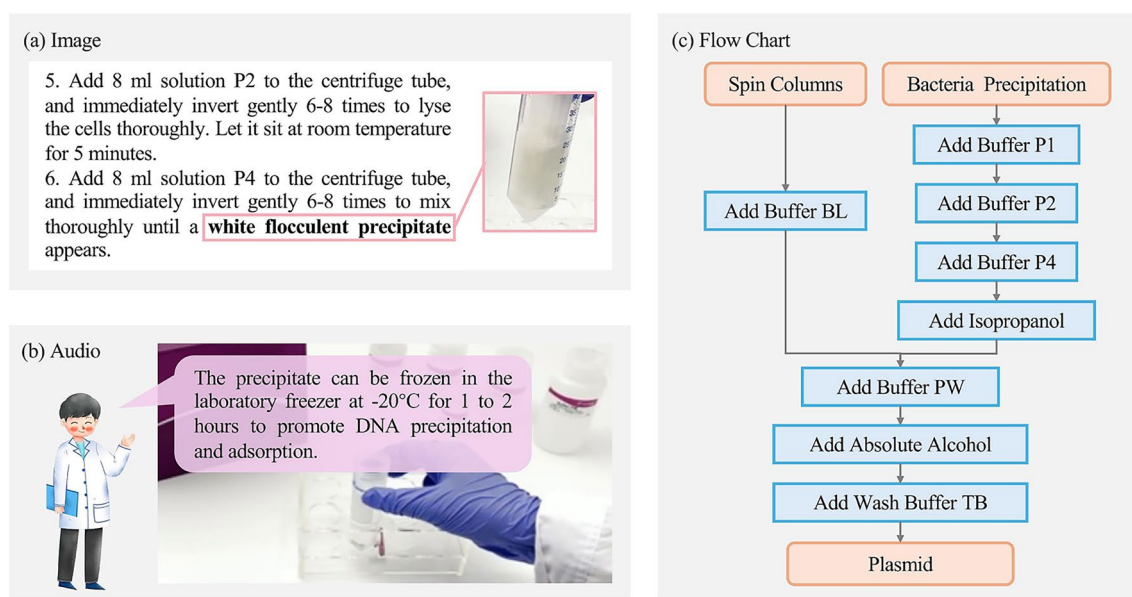


Fig. 6 Context-aware presentation of multimodal procedural guidance in AR system. **a** Integrating visual references into textual protocols, **b** providing audio commentary from experienced experimenters along-

side video tutorials, **c** using flowcharts to present phase-level overviews of experimental procedures

interactive flowcharts (Fig. 6c) can be displayed in AR prior to entering a new experimental phase to support planning and mental organization. During execution, visual references such as high-resolution images or annotated diagrams (Fig. 6a) can be surfaced at steps requiring visual verification, such as assessing solution clarity or confirming equipment setup. Audio guidance (Fig. 6b) can support moments when visual or manual attention is constrained, for instance by providing reminders, timing cues, or safety prompts. While these examples illustrate possible ways of incorporating different information formats into AR-based procedural guidance, their practical effects require further empirical investigation. In particular, future work should examine how different guidance modalities affect task efficiency, operational accuracy, and cognitive load in real laboratory workflows. Such evidence would help determine how multimodal AR guidance should balance different information forms according to the task context, rather than simply increasing the amount or variety of information presented to the user. By delivering the right form of guidance at the right moment, such systems can potentially help experimenters access and interpret procedural information more effectively during laboratory execution.

To support such context-aware presentation of procedural guidance, future AR systems could make instructional support responsive to the experimenter's ongoing activity, rather than presenting information in a fixed sequence. Prior work on context-aware immersive systems and robust AR perception suggests that user-related cues and environmental signals can support the modeling and understanding of

users' task contexts (Shen et al. 2024; Li et al. 2026). In a laboratory setting, these cues could help estimate the current protocol step through object recognition, hand-object interaction detection, or matching observed actions with predefined procedural steps. The system could then infer the experimenter's status, such as whether the user is preparing for a new phase, performing a routine task, verifying an experimental condition, hesitating, or approaching a critical safety step. For well-defined procedures, the system could apply simple rules to correlate recognized actions or objects with specific protocol steps. In more ambiguous situations, Vision Language Models (VLMs) could help interpret the first-person visual context alongside procedural instructions. Based on the identified step and execution state, the system could adapt the timing, modality, and level of detail of guidance, such as displaying a brief text reminder during routine tasks, presenting visual or video references during verification tasks, or providing audio and safety prompts when visual or manual attention is limited. In this way, context-aware presentation should not simply mean providing more information, but selectively delivering the most appropriate support based on the user's task status and the urgency of the situation.

7.1.3 Providing hierarchical representation and personalized guidance

As shown in Sect. 6.3, participants' protocol preferences varied with proficiency levels and individual strategies for understanding, recalling, and checking procedural

information. These findings indicate that a single fixed protocol view is unlikely to support all experimenters equally, motivating AR-based guidance that can organize procedural content into hierarchical structures and adapt the level of detail to users' current needs. Rather than presenting protocols as flat, linear documents, AR systems can organize experimental procedures into layered structures that allow users to access information at different levels of abstraction depending on their needs at a given moment. Procedural guidance can be decomposed into modular and indexable units corresponding to meaningful experimental stages or decision points. Importantly, selecting a single step or component can dynamically surface its associated contextual information, such as prerequisite conditions, parameter ranges, common failure modes, or linked sub-steps, enabling users to fluidly navigate related procedural knowledge without disrupting ongoing work. This design allows novice experimenters to expand steps into detailed, step-by-step instructions with embedded rationales and safety reminders, while enabling experienced experimenters to rely primarily on concise summaries or checkpoints that support efficient execution. Such hierarchical and on-demand access supports flexible navigation while helping users avoid unnecessary information processing during task execution.

Beyond static hierarchies, AR systems offer opportunities for adaptive guidance that responds to user behavior during execution. Recent advances in adaptive user interfaces suggest that information granularity and visibility can be dynamically adjusted based on observed interaction patterns (Lu et al. 2024; Shen et al. 2025). In laboratory settings, this could allow the system to surface additional guidance when users hesitate, revisit steps, or deviate from expected workflows, while remaining unobtrusive when users proceed confidently. Moreover, our findings indicate that personalization should extend beyond expertise level alone. Individual differences in learning pace, information sensitivity, and validation strategies suggest that AR-based guidance can benefit from incorporating behavioral signals such as gaze patterns (Alghofaili et al. 2019), hand movements (Li et al. 2023), and voice input (Wen et al. 2024). Leveraging these signals, future systems could adapt instructional pacing, modality emphasis, and confirmation prompts in real time, aligning guidance more closely with individual cognitive and procedural needs.

7.1.4 Enabling accessible visualization and interactive control

As shown in Sect. 6.4, participants emphasized that consulting procedural guidance during laboratory work can be inconvenient when their hands are gloved, occupied, or handling sensitive materials. They valued guidance that could

remain visible during ongoing procedures, support seamless switching between information forms, and reduce the need for direct physical interaction with external devices. These findings suggest that AR-based procedural guidance should enable accessible visualization and interactive control, allowing experimenters to consult and manage procedural information while maintaining workflow continuity. Unlike paper-based protocols, AR systems can present procedural information directly within the experimenter's workspace, enabling continuous access without disrupting manual activity or visual focus. Prior work in domains such as equipment maintenance (Nagy et al. 2024) and surgical guidance (Reinschluessel et al. 2023) demonstrated how AR can support complex, high-stakes tasks by offering hands-free, in-situ guidance. Translating these capabilities to laboratory experimentation opens opportunities to reduce workflow interruptions caused by shifting attention between physical tasks and external references.

From a design perspective, effective AR procedural guidance should combine persistent visibility with lightweight interaction mechanisms. Rather than requiring frequent manual input, systems can support voice commands, spatial anchoring, and simple gesture-based controls to allow experimenters to navigate protocol content, adjust playback, or request additional detail while remaining engaged with ongoing procedures (Eiriksdottir and Catrambone 2011). This interaction flexibility enables users to regulate information flow in response to real-time needs, such as pausing guidance during delicate steps or quickly reviewing upcoming actions. Importantly, accessible visualization in AR is not only about convenience but also about maintaining continuity of action. By keeping procedural guidance collocated with experimental activity, AR systems can support smoother transitions between steps, reduce unnecessary task switching, and help experimenters sustain situational awareness throughout complex workflows.

7.2 Applicability across diverse laboratory workflows

This study aimed to explore experimenters' needs for laboratory procedural guidance tools and to derive design opportunities for AR-based procedural guidance, using a representative molecular biology workflow as a case study. Although grounded in a specific case study, the resulting insights and proposed design opportunities may extend to a broader range of laboratory workflows and experimental contexts.

Most laboratory experiments rely on standardized protocols to ensure procedural consistency and reproducibility. During experimental execution, experimenters frequently need to consult protocol information, verify procedural

details, and coordinate instructions with ongoing physical operations. These activities create challenges related to information access, attention management, and workflow continuity. Consequently, the proposed design opportunities regarding guidance content delivery, information presentation, cognitive load considerations, and personalized user support may also be applicable to a broad range of protocol-driven laboratory workflows that require continuous interpretation and execution of procedural instructions.

In addition, the selected plasmid extraction workflow incorporates several operational activities frequently encountered in laboratory experiments, including sample preparation, reagent handling, and liquid transfer. Although different laboratory workflows may involve different samples and reagents, these activities often follow similar operational procedures and present comparable challenges, such as understanding subtle operational requirements (e.g., mixing intensity and handling techniques), interpreting the physical states of experimental materials, and accessing critical parameters such as reagent volumes and reaction times. Therefore, other laboratory workflows that involve similar operational activities and procedural requirements may also benefit from the insights and design opportunities identified in this work.

7.3 Limitation and future works

While this work offers an initial human-centered investigation into laboratory protocol practices and the design space of AR-based procedural guidance tools, several limitations should be acknowledged. In the following, we discuss these limitations and highlight several promising directions for future research to further advance the design of AR-based laboratory procedural guidance systems.

7.3.1 Participant scope

Although the study involved participants with varying levels of laboratory experience, which provided initial perspectives on protocol use, laboratory workflow challenges, and expectations for AR-based procedural guidance, the participant pool remained limited in scope. The number and diversity of participants in both the interview and participatory design sessions may constrain the generalizability and robustness of the qualitative findings and situated design feedback. The findings may not fully reflect the perspectives of broader laboratory communities, including experimenters with different professional roles, a broader range of expertise levels, and diverse disciplinary backgrounds. Future studies could involve participants from different professional roles, such as senior technicians, industrial researchers, and clinical laboratory staff; experimenters from domains such as

chemistry, materials science, cell biology, and clinical diagnostics; and participants with broader levels of professional expertise, from novice trainees to experienced researchers and senior laboratory personnel.

7.3.2 Workflow diversity and validation

This study focuses on a single molecular biology experiment as a representative case study. As discussed in Sect. 7.2, some of the insights and design opportunities identified may also apply to other protocol-driven laboratory workflows. However, biochemical experiments include a wide variety of procedures and experimental settings. While many of these share common operational activities, they can differ significantly in their specific operational requirements, task execution details, and experimental constraints. For example, some experiments primarily involve manual tasks like sample preparation and liquid handling, while others depend heavily on instrument operation, require long-duration monitoring, or involve collaboration among multiple experimenters. These variations may lead to specific needs for experimenters that were not fully addressed in the chosen case study and require further exploration. Therefore, future work should investigate a broader range of laboratory workflows, task scenarios, experimental types, and complexity levels. This additional research will help validate the applicability of the insights and design opportunities identified and will also help uncover further design considerations for AR-based procedural guidance.

7.3.3 Full-system evaluation in laboratory settings

The lightweight AR prototype served as a design probe, allowing participants to experience core AR interactions, including hands-free access, content guidance, and interactive control, and to reflect on the role of AR in laboratory procedural guidance and articulate situated design needs. However, the prototype serves as a simplified implementation and does not fully capture the performance of a comprehensive AR-based procedural guidance system in actual laboratory practice. Real-world laboratory workflows often entail stricter time constraints, heightened safety standards, longer task durations, diverse equipment, unpredictable procedural deviations, and greater demand for reliable interaction. Consequently, future research should focus on developing higher-fidelity, fully integrated AR systems and evaluating them under realistic, complex, time-sensitive, and safety-critical laboratory conditions. Subsequent full-system evaluations should quantitatively assess system-level outcomes such as usability, task completion time, error rates, workflow disruptions, safety-related issues, cognitive workload, interaction reliability, and long-term integration

into routine laboratory operations under realistic and high-pressure laboratory conditions.

7.3.4 Diverse laboratory environments

An interesting direction for future research is to investigate how AR-based procedural guidance can support laboratory work across diverse laboratory environments. While the current study focused on a single laboratory environment to enable a focused investigation of experimenters' needs and challenges, different laboratory environments may place different demands on procedural guidance. For example, educational laboratories often involve novice users who may benefit from detailed guidance and error-prevention mechanisms, whereas research laboratories may require more flexible access to protocol information and experimental knowledge. Industrial laboratories may further emphasize standardization, compliance, and traceability throughout experimental procedures. Future work could examine how the identified design opportunities should be adapted to support these differing requirements. Such investigations may provide deeper insights into how AR-based procedural guidance systems can balance guidance, flexibility, and standardization across diverse laboratory environments.

7.3.5 Cross-platform procedural guidance

Future procedural guidance systems should support seamless integration across multiple platforms, including desktop computers, mobile devices, wearables, and AR headsets. Experimenters frequently transition between devices at different stages of experimental work, and maintaining continuity of procedural information across platforms remains a challenge. This calls for adaptive visualization strategies that preserve information structure while optimizing presentation for different display constraints. Context-aware handoff mechanisms could enable experimenters to switch devices fluidly without losing procedural context. Cloud-based synchronization and edge computing may further support low-latency access, ensuring responsiveness in time-critical laboratory scenarios.

7.3.6 Real-time and adaptive experimental support

As discussed in Sect. 7.1.2, AR-based systems could support real-time adaptive guidance by using computer vision and multimodal sensing to infer user actions, experimental progress, and potential procedural deviations. Such feedback should be carefully designed to remain informative but unobtrusive, with its granularity and modality adapted to user expertise and task criticality. While context-aware presentation is a promising direction for adaptive procedural

guidance, wet-lab environments introduce several practical challenges that may affect its technical feasibility. Reflective surfaces, transparent containers, visually similar reagents, glove occlusion, splashes, and contamination may affect the reliability of computer vision, hand-object interaction detection, and eye tracking. As a result, current perception accuracy may not yet be sufficient to support fully automated real-time context-aware guidance in complex laboratory workflows. In the short term, future systems may rely on more controlled and user-confirmed mechanisms, such as explicit step confirmation, voice-based navigation, timer-based reminders, barcode- or QR-code-based object identification, and rule-based prompts. In the longer term, with the development of multi-view sensing, higher-accuracy recognition models, improved sensors, and more reliable reasoning methods, these context-aware processes may be gradually automated. Moreover, their deployment in wet-lab workflows will require systematic evaluation of perception accuracy, failure modes, and safety implications under realistic laboratory conditions.

7.3.7 Longitudinal and ecological evaluation

Evaluating the long-term impact of AR-based procedural guidance remains an open challenge. Future longitudinal studies should examine how experimenters' preferences, interaction strategies, and trust in AR-based guidance evolve over time, and how these changes influence sustained adoption in routine laboratory workflows. For example, while experimenters may initially value the novelty and accessibility of AR-based guidance, continued use may depend more on whether the system remains reliable, minimizes workflow interruptions, and fits naturally into repeated laboratory routines. Beyond short-term usability and performance gains, future work should examine how sustained use of intelligent guidance tools influences skill acquisition, procedural understanding, and experimental autonomy over time. Longitudinal studies conducted in real laboratory settings could assess effects on productivity, reproducibility, and knowledge transfer. Potential risks such as over-reliance on automation or reduced critical engagement with protocols must be examined to ensure that intelligent assistance supports, rather than undermines, scientific reasoning and rigor.

8 Conclusion

This research presents a two-stage study that explores experimenters' visions and expectations for AR-based procedural guidance tools to support more efficient experimental workflows. We first conducted semi-structured interviews

to uncover current practices and key challenges associated with existing laboratory protocols, which subsequently informed the design of AR prototype probe. The participatory design sessions that followed provided deeper insights into experimenters' situated procedural information needs during hands-on experimentation, as well as how laboratory protocols should be designed to better align with their expectations. Our findings suggest that procedural guidance tools should offer comprehensive and reliable content, clear and well-structured organization, multimodal representations, adaptability to experimenters' expertise levels and cognitive preferences, and accessible, interactive visualizations. Drawing on these insights, together with cognitive theories and prior work, we articulate a set of design implications for developing more user-centered AR-based procedural guidance tools. By foregrounding experimenters' perspectives, this work bridges Human-Computer Interaction (HCI) research and scientific practice, demonstrating how user-centered design can contribute to more effective, usable, and reproducible laboratory procedural guidance tools.

Acknowledgements This work was supported in part by the National Science and Technology Major Project (2022ZD0114900), the State Key Laboratory of General Artificial Intelligence (SKL-AGI2024OP16).

Author contributions Yuxin Shen and Wei Liang collected the data. Jizhan Ma contributed to the methodology. Yuxin Shen and Wei Liang performed the analysis. All authors contributed to the interpretation of the results and the writing of the manuscript. Yixin Zhu revised the manuscript for important intellectual content. All authors read and approved the final manuscript.

Data availability The data that support the findings of this study are not publicly available due to privacy and ethical restrictions involving human participants, but are available from the corresponding author upon reasonable request.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest relevant to this article.

Ethical approval This study was approved by the institutional research ethics committee of the authors' institution.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication Participants provided consent for the publication of anonymized data and materials.

References

- Agrawal, R., Gollapudi, S., Kannan, A.: Enriching textbooks with images. In: Proceedings of International Conference on Information and Knowledge Management (CIKM), pp. 1847–1856. ACM (2011). <https://doi.org/10.1145/2063576.2063843>
- Alghofaili, R., Sawahata, Y., Huang, H.: Lost in style: gaze-driven adaptive aid for VR navigation. In: ACM Conference on Human Factors in Computing Systems (CHI), pp. 1–12. ACM (2019). <https://doi.org/10.1145/3290605.3300578>
- Al-Thanyyan, S.S., Azmi, A.M.: Automated text simplification: a survey. *ACM Comput. Surv. (CSUR)* **54**(2), 1–36 (2021). <https://doi.org/10.1145/3442695>
- Alves, J., Marques, B., Oliveira, M.: Comparing spatial and mobile augmented reality for guiding assembling procedures with task validation. In: IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC), pp. 1–6. IEEE (2019). <https://doi.org/10.1109/ICARSC.2019.8733642>
- Anwer, M.O., Zagrouba, E.: Optimized design and evaluation of immersive multiuser virtual reality educational applications. *CCF Trans. Pervasive Comput. Interact.* (2025). <https://doi.org/10.1007/s42486-025-00217-9>
- Baek, M., DiMaio, F., Anishchenko, I., et al.: Accurate prediction of protein structures and interactions using a three-track neural network. *Science* **373**(6557), 871–876 (2021). <https://doi.org/10.1126/science.abj8754>
- Baker, M.: 1,500 scientists lift the lid on reproducibility. *Nature* **533**(7604), 452–454 (2016). <https://doi.org/10.1038/533452a>
- Bhardwaj, A., Kishore, S., Pandey, D.K.: Artificial intelligence in biological sciences. *Life* **12**(9), 1430 (2022). <https://doi.org/10.3390/life12091430>
- Braun, V., Clarke, V.: Using thematic analysis in psychology. *Qual. Res. Psychol.* **3**(2), 77–101 (2006). <https://doi.org/10.1191/1478088706qp063oa>
- Buchner, J., Buntins, K., Kerres, M.: The impact of augmented reality on cognitive load and performance: a systematic review. *J. Comput. Assist. Learn.* **38**(1), 285–303 (2022). <https://doi.org/10.1111/jcal.12617>
- Carroll, J.A., Minnen, G., Pearce, D., et al.: Simplifying text for language-impaired readers. In: Proceedings of Conference of the European Chapter of the Association for Computational Linguistics (EACL), pp. 269–270. Association for Computational Linguistics (1999). <https://aclanthology.org/E99-1042>
- Castelo, S., Rulff, J., McGowan, E., et al.: ARGUS: visualization of AI-assisted task guidance in AR. *IEEE Trans. Vis. Comput. Graph (TVCG)* **30**(01), 1313–1323 (2024). <https://doi.org/10.1109/TVCG.2023.3327396>
- Chandler, P., Sweller, J.: Cognitive load theory and the format of instruction. *Cogn. Instr.* **8**(4), 293–332 (1991). https://doi.org/10.1207/s1532690xcio804_2
- Chen, C., Nguyen, C., Hoffswell, J.: PaperToPlace: transforming instruction documents into spatialized and context-aware mixed reality experiences. In: ACM Symposium on User Interface Software and Technology (UIST), pp. 1–21. ACM (2023). <https://doi.org/10.1145/3586183.3606832>
- Chi, P., Frey, N., Panovich, K.: Automatic instructional video creation from a markdown-formatted tutorial. In: ACM Symposium on User Interface Software and Technology (UIST), pp. 677–690. ACM (2021). <https://doi.org/10.1145/3472749.3474778>
- Cui, J., Gong, Z., Jia, B., et al.: ProBio: a protocol-guided multimodal dataset for molecular biology lab. In: Proceedings of Advances in Neural Information Processing Systems (NeurIPS), pp. 41543–41571. Curran Associates Inc. (2024). <https://doi.org/10.48550/arXiv.2311.00556>

- Dannenfelser, R., Zhong, J., Zhang, R., et al.: Into the single cell multiverse: an end-to-end dataset for procedural knowledge extraction in biomedical texts. In: Proceedings of Advances in Neural Information Processing Systems (NeurIPS), pp. 10922–10934. Curran Associates Inc. (2023). <https://doi.org/10.48550/arXiv.2309.01812>
- Deshpande, A., Kim, I.: The effects of augmented reality on improving spatial problem solving for object assembly. *Adv. Eng. Inform.* **38**, 760–775 (2018). <https://doi.org/10.1016/j.aei.2018.10.004>
- Eiriksdottir, E., Catrambone, R.: Procedural instructions, principles, and examples: how to structure instructions for procedural tasks to enhance performance, learning, and transfer. *Hum. Factors* **53**(6), 749–770 (2011). <https://doi.org/10.1177/0018720811419154>
- Eom S, Dickinson, M., Sachs, R.: Sense-specific lexical information for reading assistance. In: Proceedings of the Seventh Workshop on Building Educational Applications Using NLP, pp. 316–325. Association for Computational Linguistics (2012). <https://aclanthology.org/W12-2038>
- Feinberg, S., Murphy, M.: Applying cognitive load theory to the design of web-based instruction. In: Proceedings of IEEE Professional Communication Society International Professional Communication Conference and Proceedings of the 18th Annual ACM International Conference on Computer Documentation: Technology & Teamwork, pp. 353–360. IEEE Educational Activities Department (2000). <https://doi.org/10.1109/IPCC.2000.887293>
- Fiorentino, M., Uva, A.E., Gattullo, M., et al.: Augmented reality on large screen for interactive maintenance instructions. *Comput. Ind.* **65**(2), 270–278 (2014). <https://doi.org/10.1016/j.compind.2013.11.004>
- Funk, M., Kosch, T., Schmidt, A.: Interactive worker assistance: comparing the effects of in-situ projection, head-mounted displays, tablet, and paper instructions. In: Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing, pp. 934–939. ACM (2016). <https://doi.org/10.1145/2971648.2971706>
- Gautier, L.: Reproducibility in the lab: the proof is in the protocol. *Biotechniques* **73**(2), 71–74 (2022). <https://doi.org/10.2144/btn-2022-0074>
- Govindasamy, T.: Successful implementation of e-learning: pedagogical considerations. *Internet Higher Educ.* **4**(3–4), 287–299 (2001). [https://doi.org/10.1016/S1096-7516\(01\)00071-9](https://doi.org/10.1016/S1096-7516(01)00071-9)
- Haddaway, N.R., Verhoeven, J.T.: Poor methodological detail precludes experimental repeatability and hampers synthesis in ecology. *Ecol. Evol.* **5**(19), 4451–4454 (2015). <https://doi.org/10.1002/ece3.1722>
- Hayward, G.: Usability of consumer product self-assembly instructions—a historical review of ergonomics research to create an international standard. *Ergonomics* **68**(6), 893–912 (2025). <https://doi.org/10.1080/00140139.2024.2377699>
- Hinkelmann, K., Kempthorne, O.: Design and Analysis of Experiments, Volume 1: Introduction to experimental design, vol. 1. Wiley, New York (2007)
- Hou, L., Wang, X., Bernold, L., et al.: Using animated augmented reality to cognitively guide assembly. *J. Comput. Civ. Eng.* **27**(5), 439–451 (2013). [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000184](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000184)
- Huang, G., Qian, X., Wang, T.: AdapTutAR: an adaptive tutoring system for machine tasks in augmented reality. In: ACM Conference on Human Factors in Computing Systems (CHI), ACM (2021). <https://doi.org/10.1145/3411764.3445283>
- Ioannidis, J.P.: How to make more published research true. *Rev. Cubana Inf. Cienc. Salud (ACIMED)* **26**(2), 187–200 (2015). <https://doi.org/10.1371/journal.pmed.1001747>
- Jiang, S., Evans-Yamamoto, D., Bersenev, D., et al.: ProtoCode: leveraging large language models (LLMs) for automated generation of machine-readable PCR protocols from scientific publications. *SLAS Technol.* **29**(3), 100134 (2024). <https://doi.org/10.1016/j.slast.2024.100134>
- Keelawat, P., Suzuki, R.: Transforming procedural instructions into in-situ augmented reality guides with InstructAR. In: ACM Symposium on User Interface Software and Technology (UIST), ACM (2024). <https://doi.org/10.1145/3672539.3686321>
- Knoop-van Campen, C.A., Segers, E., Verhoeven, L.: The modality and redundancy effects in multimedia learning in children with dyslexia. *Dyslexia* **24**(2), 140–155 (2018). <https://doi.org/10.1002/dys.1585>
- Lai, A.F., Chen, C.H., Lee, G.Y.: An augmented reality-based learning approach to enhancing students' science reading performances from the perspective of the cognitive load theory. *Br. J. Edu. Technol.* **50**(1), 232–247 (2019). <https://doi.org/10.1111/bj et.12716>
- Lampen, E., Teuber, J., Gaisbauer, F., et al.: Combining simulation and augmented reality methods for enhanced worker assistance in manual assembly. *Proc. Cirp* **81**, 588–593 (2019). <https://doi.org/10.1016/j.procir.2019.03.160>
- Lazar, J., Feng, J.H., Hochheiser, H.: Research Methods in Human-Computer Interaction, 2nd edn. Elsevier Science, Amsterdam (2017)
- Leake, M., Shin, H.V., Kim, J.O.: Generating audio-visual slideshows from text articles using word concreteness. In: ACM Conference on Human Factors in Computing Systems (CHI), pp. 1–11. ACM (2020). <https://doi.org/10.1145/3313831.3376519>
- Leroy, G., Endicott, J.E., Kauchak, D., et al.: User evaluation of the effects of a text simplification algorithm using term familiarity on perception, understanding, learning, and information retention. *J. Med. Internet Res.* **15**(7), e2569 (2013). <https://doi.org/10.2196/jmir.2569>
- Li, Z., Liang, C., Wang, Y., et al.: Enabling voice-accompanying hand-to-face gesture recognition with cross-device sensing. In: ACM Conference on Human Factors in Computing Systems (CHI), pp. 1–17. ACM (2023). <https://doi.org/10.1145/3544548.3581008>
- Li, B., Guo, X., Zhao, Y., et al.: Dynolayout: robust layout estimation from event-stream for extended reality under dynamic scenarios. *CCF Trans. Pervasive Comput. Interact.* **8**(1), 149–164 (2026). <https://doi.org/10.1007/s42486-025-00212-0>
- Liu, Y., Lv, L., Zhang, X.: BioProBench: Comprehensive Dataset and Benchmark in Biological Protocol Understanding and Reasoning (2025). [arXiv:2505.07889](https://arxiv.org/abs/2505.07889) arXiv preprint
- Lu, F., Chen, M., Hsu, H.: Adaptive 3D UI placement in mixed reality using deep reinforcement learning. In: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems, ACM (2024). <https://doi.org/10.1145/3613905.3651059>
- Macleod, M.R., Michie, S., Roberts, I., et al.: Biomedical research: increasing value, reducing waste. *Lancet* **383**(9912), 101–104 (2014). [https://doi.org/10.1016/S0140-6736\(13\)62329-6](https://doi.org/10.1016/S0140-6736(13)62329-6)
- Nagra, N.S., Bleys, J., Champagne, D., et al.: Six factors affecting reproducibility in life science research and how to handle them. (2019). <https://www.nature.com/articles/d42473-019-00004-y>
- Nagy, A., Spyridis, Y., Mills, G.J.: User experience evaluation of AR assisted industrial maintenance and support applications. In: 2024 10th International Conference on Virtual Reality (ICVR), pp. 270–277. IEEE (2024). <https://doi.org/10.1109/ICVR62393.2024.10869099>
- Nisioi, S., Štajner, S., Ponzetto, S.P.: Exploring neural text simplification models. In: Annual Meeting of the Association for Computational Linguistics (ACL), pp. 85–91. Association for Computational Linguistics (2017). <https://doi.org/10.18653/v1/P17-2014>
- Obermair, F., Althaler, J., Seiler, U.: Maintenance with augmented reality remote support in comparison to paper-based instructions: experiment and analysis. In: International Conference on Industrial Engineering and Applications (ICIEA), pp. 942–947.

- Institute of Electrical and Electronics Engineers Inc (2020). <https://doi.org/10.1109/ICIEA49774.2020.9102078>
- Ohsugi, T., Matsushita, M.: Investigation of recipe components to resolve ambiguity in cooking procedure. In: Proceedings of the 6th Asian Conference on Information Systems, pp. 68–73 (2017). <http://id.nii.ac.jp/1457/00003718/>
- O'Shaughnessy, J.: Consumer Behaviour: Perspectives, Findings and Explanations. Bloomsbury Academic, London (2012)
- Ouyang, Y., Shen, L., Wang, Y.: NotePlayer: engaging computational notebooks for dynamic presentation of analytical processes. In: ACM Symposium on User Interface Software and Technology (UIST), pp. 1–20. ACM (2024). <https://doi.org/10.1145/365477.73676410>
- Plant, A., Hanisch, R.: Reproducibility in science: a metrology perspective. *Harv. Data Sci. Rev.* **2**(4), 26 (2020). <https://doi.org/10.1162/99608f92.eb6dde4>
- Polvi, J., Taketomi, T., Moteki, A., et al.: Handheld guides in inspection tasks: augmented reality versus picture. *IEEE Trans. Vis. Comput. Graph (TVCG)* **24**(7), 2118–2128 (2017). <https://doi.org/10.1109/TVCG.2017.2709746>
- Qu, Y., Huang, K., Yin, M., et al.: CRISPR-GPT for agentic automation of gene-editing experiments. *Nat. Biomed. Eng.* (2025). <https://doi.org/10.1038/s41551-025-01463-z>
- Rahal, R.M., Hamann, H., Brohmer, H.: Sharing the recipe: reproducibility and replicability in research across disciplines. *Res. Ideas Outcomes RIO* **8**, 1–20 (Article e89980) (2022). <https://doi.org/10.3897/rio.8.e89980>
- Reinschluessel, A.V., Muender, T., Fischer, R., et al.: Versatile immersive virtual and augmented tangible or—using VR, AR and tangibles to support surgical practice. In: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems, pp. 1–5. ACM (2023). <https://doi.org/10.1145/3544549.3583895>
- Ren, H., Wu, M., Croisdale, G.T.: Rubikon: intelligent tutoring for rubik's cube learning through AR-enabled physical task reconfiguration. In: ACM SIGCHI Conference on Designing Interactive Systems (DIS), pp. 3549–3562. ACM (2025). <https://doi.org/10.1145/3715336.3735788>
- Roohani, Y.H., Vora, J., Huang, Q., et al.: BioDiscoveryAgent: an AI agent for designing genetic perturbation experiments. In: ICLR 2024 Workshop on Large Language Model (LLM) Agents. <https://openreview.net/forum?id=Ve2qQWQc4D> (2024)
- Scarton, C., Aprosio, A.P., Tonelli, S., et al.: MUSST: a multilingual syntactic simplification tool. In: Proceedings of the IJCNLP 2017, System Demonstrations, pp. 25–28. Association for Computational Linguistics <https://aclanthology.org/117-3007> (2017)
- Selwyn, M.R.: Principles of Experimental Design for the Life Sciences. CRC Press, Boca Raton (1996)
- Semeraro, A., Turmo Vidal, L.: Visualizing instructions for physical training: exploring visual cues to support movement learning from instructional videos. In: ACM Conference on Human Factors in Computing Systems (CHI). ACM (2022). <https://doi.org/10.1145/3491102.3517735>
- Shan, S., Luo, S., Yang, Z., et al.: Deep learning guided optimization of human antibody against SARS-CoV-2 variants with broad neutralization. *Proc. Natl. Acad. Sci. (PNAS)* **119**(11), e2122954119 (2022). <https://doi.org/10.1073/pnas.2122954119>
- Shen, Y., Xu, M., Liang, W.: Context-aware head-and-eye motion generation with diffusion model. In: 2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR), pp. 157–167. IEEE (2024)
- Shen, Y., Liang, W., Ma, J.: LiteAT: a data-lightweight and user-adaptive VR telepresence system for remote education. *IEEE Trans. Vis. Comput. Graph.* **31**(11), 9570–9580 (2025). <https://doi.org/10.1109/tvcg.2025.3616747>
- Shi, J., Jain, R., Chi, S.: CARING-AI: towards authoring context-aware augmented reality instruction through generative artificial intelligence. In: ACM Conference on Human Factors in Computing Systems (CHI), ACM (2025). <https://doi.org/10.1145/53706598.3713348>
- Siddharthan, A.: Syntactic simplification and text cohesion. *Res. Lang. Comput.* **4**, 77–109 (2006). <https://doi.org/10.1007/s11168-006-9011-1>
- Siddharthan, A.: A survey of research on text simplification. *Int. J. Appl. Linguist.* **165**(2), 259–298 (2014). <https://doi.org/10.1075/itl.165.2.06sid>
- Wang, C.H., Lo, W.J., Wang, M.J.J.: Usability evaluation of augmented reality-based maintenance instruction system. *Hum. Factors Ergon. Manuf. Serv. Ind.* **32**(3), 239–255 (2022). <https://doi.org/10.1002/hfm.20942>
- Weerasinghe, M., Quigley, A., Pucihar, K.Č, et al.: Arigatō: effects of adaptive guidance on engagement and performance in augmented reality learning environments. *IEEE Trans. Vis. Comput. Graph (TVCG)* **28**(11), 3737–3747 (2022). <https://doi.org/10.1109/tvcg.2022.3203088>
- Wen, S., Ping, S., Wang, J.: AdaptiveVoice: cognitively adaptive voice interface for driving assistance. In: ACM Conference on Human Factors in Computing Systems (CHI), ACM (2024). <https://doi.org/10.1145/3613904.3642876>
- Wu, G., Qian, J., Castelo Quispe, S.: ARTiST: automated text simplification for task guidance in augmented reality. In: ACM Conference on Human Factors in Computing Systems (CHI), ACM (2024). <https://doi.org/10.1145/3613904.3642772>
- Xia, H., Jacobs, J., Agrawala, M.: Crosscast: adding visuals to audio travel podcasts. In: ACM Symposium on User Interface Software and Technology (UIST), pp. 735–746. ACM (2020). <https://doi.org/10.1145/3379337.3415882>
- Yang, Z., Shi, J., Jiang, W., et al.: Influences of augmented reality assistance on performance and cognitive loads in different stages of assembly task. *Front. Psychol.* **10**, 458057 (2019). <https://doi.org/10.3389/fpsyg.2019.01703>
- Zhai, K.y., Cao, Y.m., Hou, W.j., et al.: Interactive mixed reality cooking assistant for unskilled operating scenario. In: International Conference on Human-Computer Interaction, pp. 178–195. Springer (2020). https://doi.org/10.1007/978-3-030-49698-2_13
- Zhang, N., Yan, Y., Suzuki, R.: From following to understanding: investigating the role of reflective prompts in AR-guided tasks to promote user understanding. In: ACM Conference on Human Factors in Computing Systems (CHI), ACM (2025). <https://doi.org/10.1145/3706598.3713293>
- Zhao, A.Y., Gunturu, A., Do, E.Y.L.: Guided reality: generating visually-enriched AR task guidance with LLMs and vision models. In: ACM Symposium on User Interface Software and Technology (UIST), pp. 1–15. ACM (2025). <https://doi.org/10.1145/374605.93747784>
- Zhong, M., Li, G., Chi, P.: HelpViz: automatic generation of contextual visual mobile tutorials from text-based instructions. In: ACM Symposium on User Interface Software and Technology (UIST), pp. 1144–1153. ACM (2021). <https://doi.org/10.1145/3472749.3474812>
- Zhu, Z., Liu, Z., Zhang, Y.: LearnIoTVR: an end-to-end virtual reality environment providing authentic learning experiences for internet of things. In: ACM Conference on Human Factors in Computing Systems (CHI), ACM (2023). <https://doi.org/10.1145/3544548.3581396>

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.



Yuxin Shen Yuxin Shen is a Ph.D. student in the School of Computer Science and Technology at Beijing Institute of Technology (BIT), supervised by Prof. Wei Liang. He is affiliated with the Perception, Interaction, and Embodiment Lab. His research interests include Human-Computer Interaction (HCI), eXtended Reality (XR), and intelligent interactive systems. His work received a Best Paper Honorable Mention at IEEE ISMAR.



Yixin Zhu Yixin Zhu received the Ph.D. degree in statistics from the University of California, Los Angeles (UCLA), in 2018. He is currently an Assistant Professor with Peking University, Beijing, China, jointly appointed in the School of Psychological and Cognitive Sciences and the Institute for Artificial Intelligence. His research interests include cognitive reasoning, embodied intelligence, computer vision, and human-centric AI, with a particular emphasis on integrating com-

monsense knowledge with multimodal perception to build interactive intelligent systems. He has published in Science, Nature Machine Intelligence, Nature Human Behaviour, Science Robotics, and Science Advances, among others.



Wei Liang Wei Liang is a professor at the School of Computer Science and Technology at Beijing Institute of Technology (BIT), China. She earned her Ph.D. in Computer Science from BIT in 2005. In 2014, she was a visiting professor at the Vision Cognition Learning and Autonomy (VCLA) Lab at UCLA, USA. Her research interests focus on Human-Computer Interaction (HCI) and Artificial Intelligence (AI).



Jianzhu Ma Jianzhu Ma is an Associate Professor and doctoral supervisor with the Department of Electronic Engineering and the Institute for Artificial Intelligence at Tsinghua University. He was selected for the Overseas Excellent Young Scientists Program. He previously served as a Walther Assistant Professor in the Department of Computer Science at Purdue University, with a joint appointment as an Assistant Professor in the Department of Biochemistry at Purdue University.

His research interests include artificial intelligence, systems biology, and biopharmaceuticals. He has received several awards, including the Best Paper Award at RECOMB, a leading international conference in computational biology; the Warren DeLano Award at ISMB; the Best Poster Award at the International Conference on RNA and Protein Folding; the Poster Award at the First International Conference on Urban Science; the 2024 Top Ten Advances in Bioinformatics in China; and a Best Paper Honorable Mention at ISMAR, a leading international conference in mixed and augmented reality. His research has been published in journals such as Nature, Cell, Cancer Cell, Nature Machine Intelligence, Nature Methods, Nature Cancer, Nature Biomedical Engineering, and Immunity. His papers have also been featured as cover articles in Nature Methods, Nature Machine Intelligence, and Cell Systems. In addition to his contributions to biopharmaceutical research, Dr. Ma has published more than 50 papers in leading international artificial intelligence conferences, and his work has received over 10,000 citations on Google Scholar.